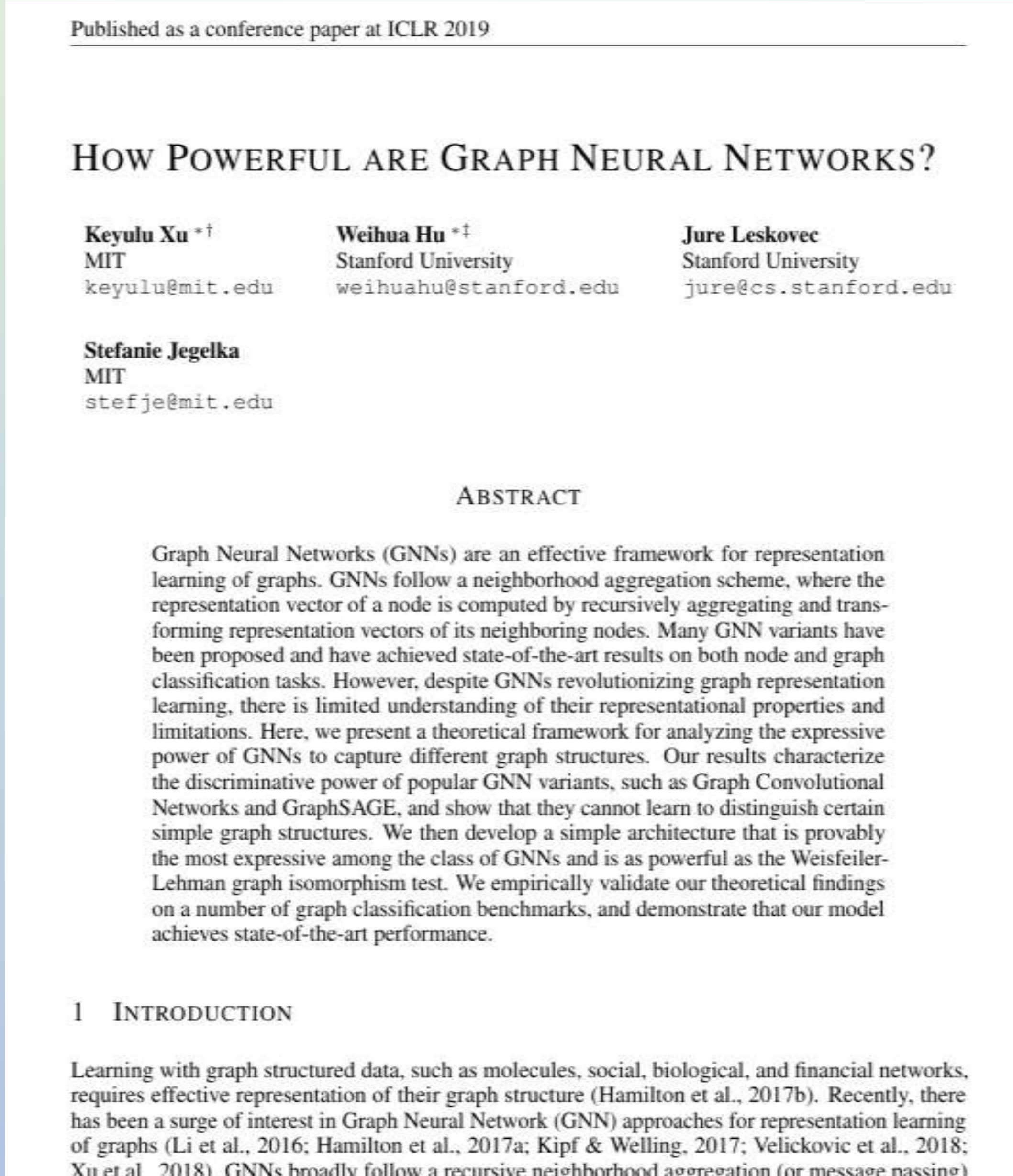
IIT DELHI MISN LAB RESEARCH INTERNSHIP

Research Paper
Presentation on :-

“HOW POWERFUL ARE GRAPH NEURAL NETWORKS ?”

By Keyulu Xu (MIT)



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HOW POWERFUL ARE GRAPH NEURAL NETWORKS?

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ABSTRACT

Graph Neural Networks (GNNs) are an effective framework for representation learning of graphs. GNNs follow a neighborhood aggregation scheme, where the representation vector of a node is computed by recursively aggregating and transforming representation vectors of its neighboring nodes. Many GNN variants have been proposed and have achieved state-of-the-art results on both node and graph classification tasks. However, despite GNNs revolutionizing graph representation learning, there is limited understanding of their representational properties and limitations. Here, we present a theoretical framework for analyzing the expressive power of GNNs to capture different graph structures. Our results characterize the discriminative power of popular GNN variants, such as Graph Convolutional Networks and GraphSAGE, and show that they cannot learn to distinguish certain simple graph structures. We then develop a simple architecture that is provably the most expressive among the class of GNNs and is as powerful as the Weisfeiler-Lehman graph isomorphism test. We empirically validate our theoretical findings on a number of graph classification benchmarks, and demonstrate that our model achieves state-of-the-art performance.

1 INTRODUCTION

Learning with graph structured data, such as molecules, social, biological, and financial networks, requires effective representation of their graph structure (Hamilton et al., 2017b). Recently, there has been a surge of interest in Graph Neural Network (GNN) approaches for representation learning of graphs (Li et al., 2016; Hamilton et al., 2017a; Kipf & Welling, 2017; Velickovic et al., 2018; Xu et al., 2018). GNNs broadly follow a recursive neighborhood aggregation (or message passing)

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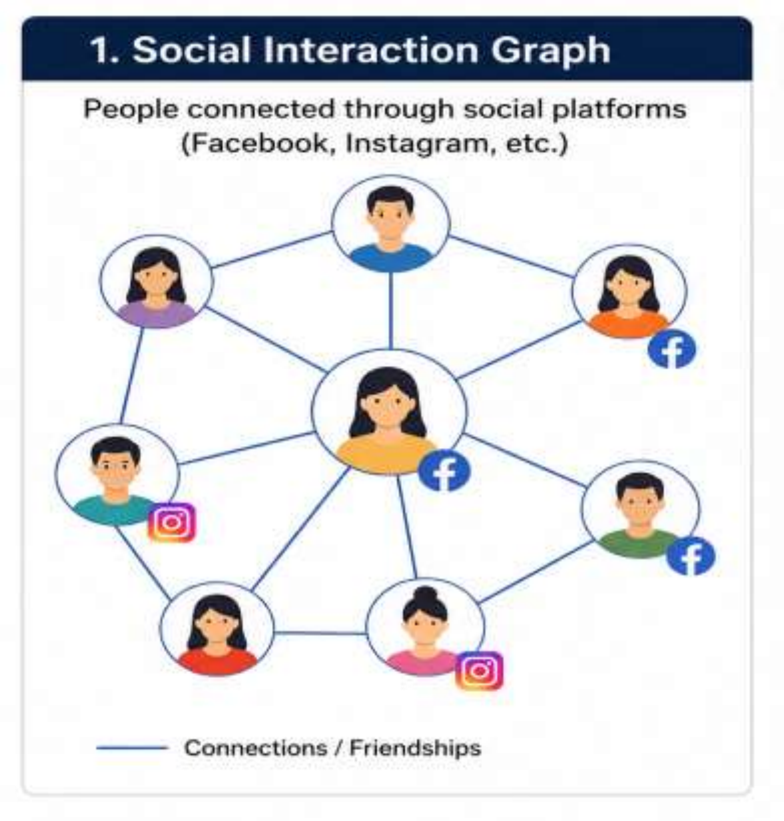
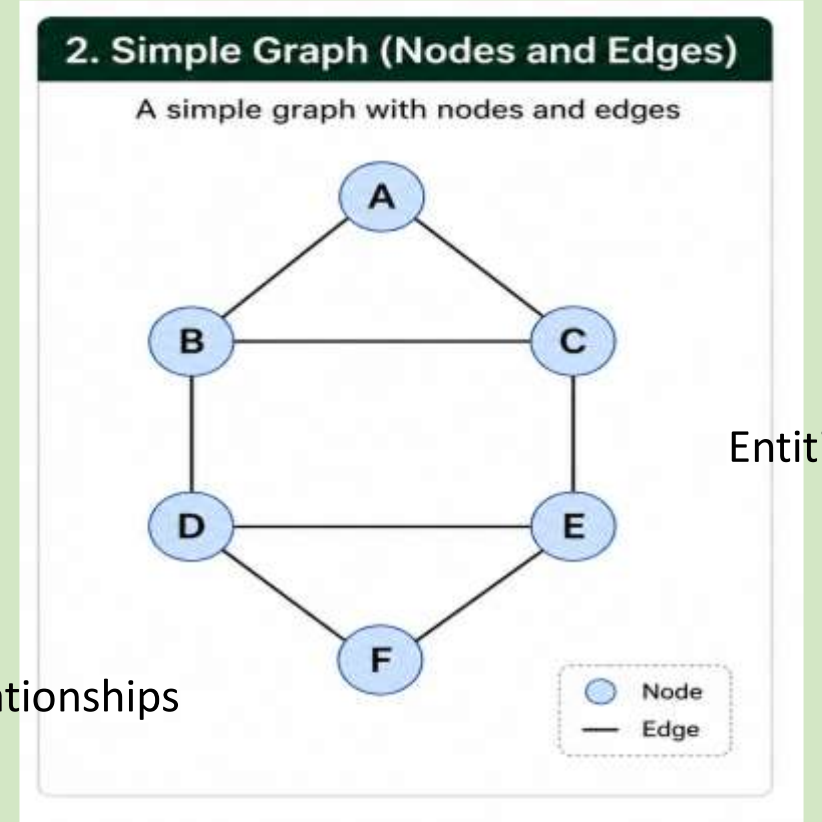
What is a GNN?





Entities

Relationships

Graph Neural Network (GNN)

- A graph consists of **Nodes + Edges**
- Nodes = entities
- Edges = relationships
- GNNs learn from a node **and its neighbouring nodes**
- Repeated neighbour aggregation helps capture graph structure

What is the Problem?

Research Problem

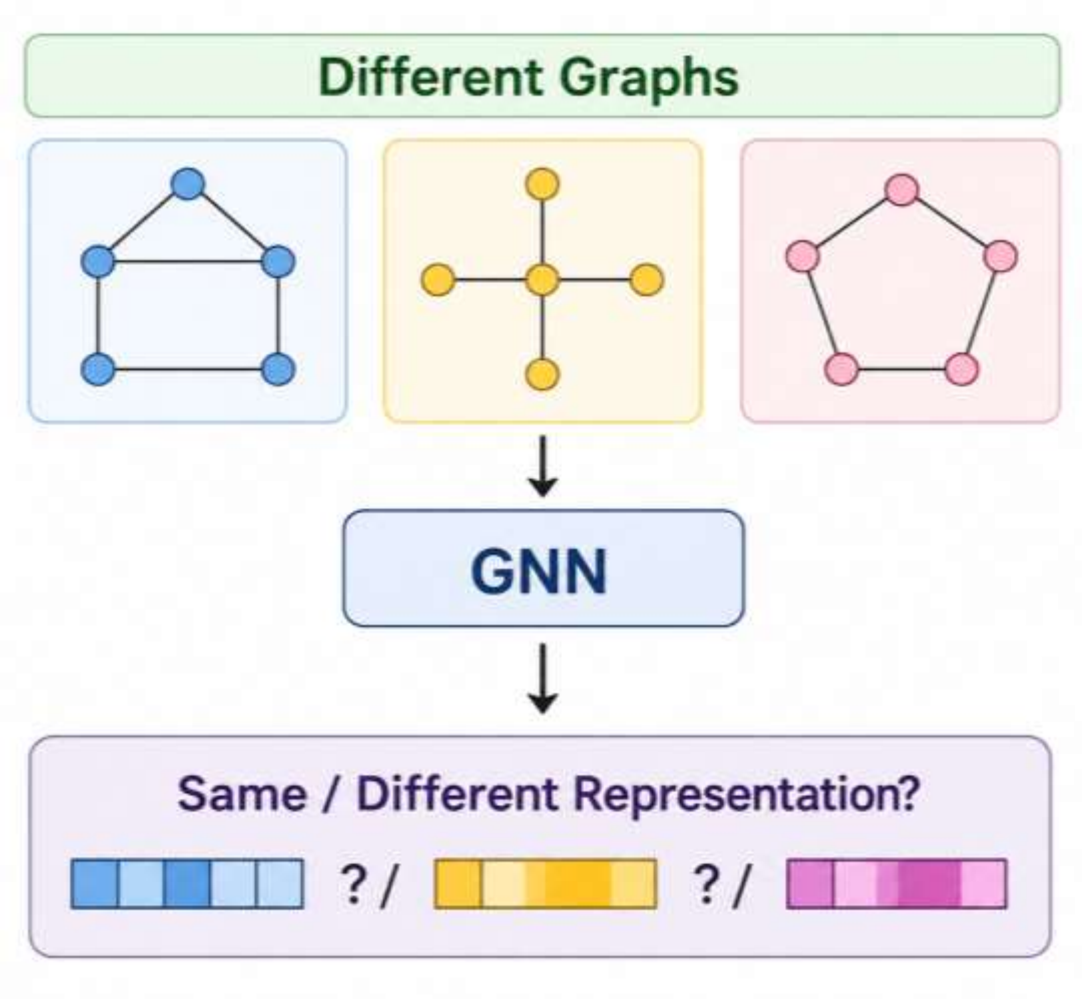
- Existing GNNs use different ways to aggregate information from neighbours:
- GCN → Mean
  GraphSAGE → Max
  Other GNNs → Different aggregation methods

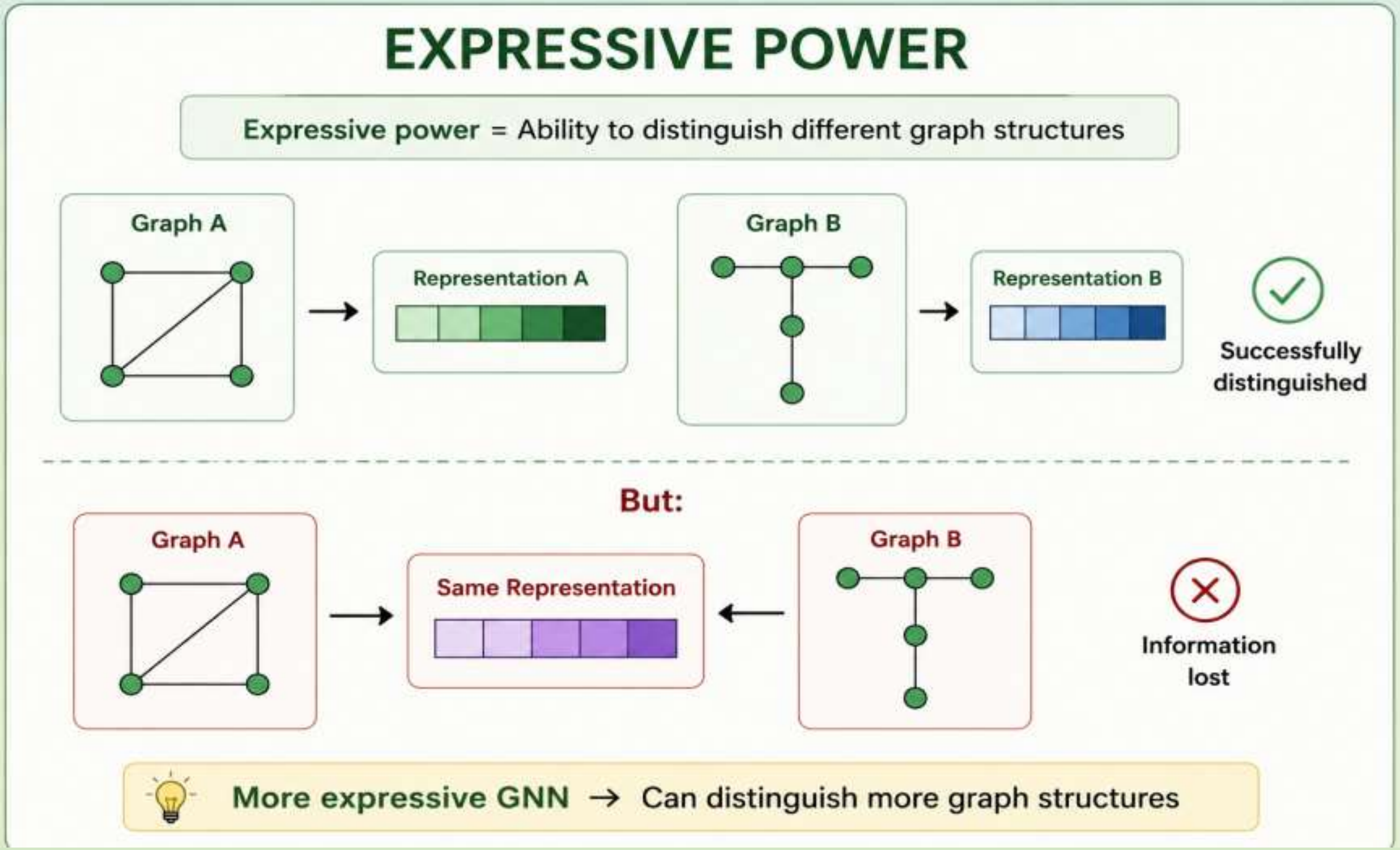
Major issue arise

- Some aggregation methods lose important structural information.

Main Research Question

- How powerful are GNNs at distinguishing different graph structures?

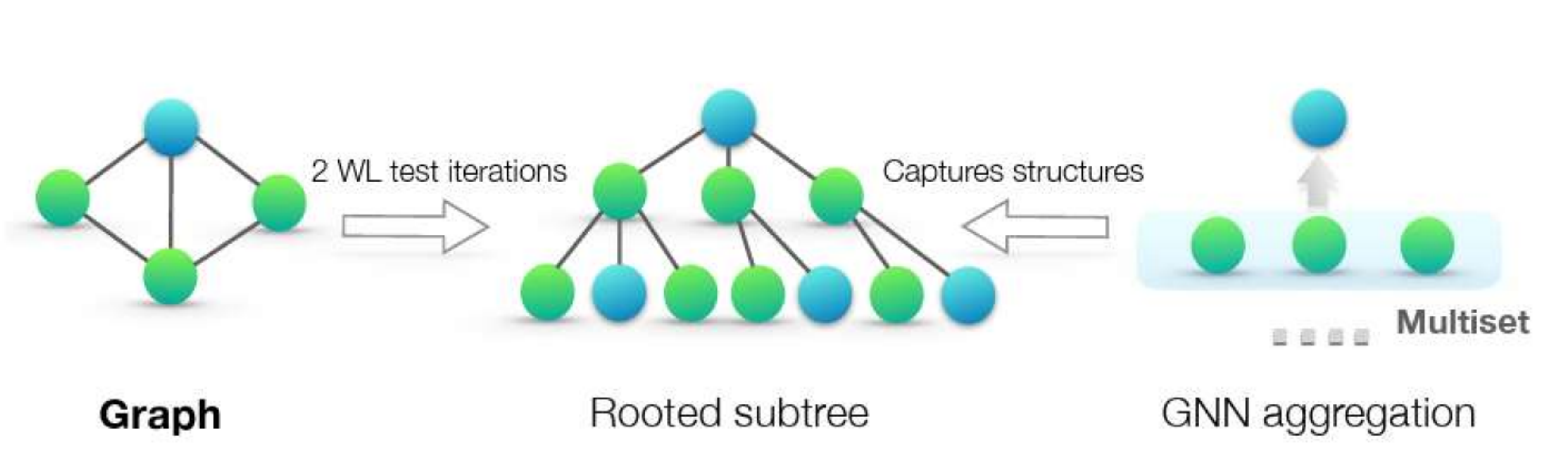




- Expressive power means the ability of a GNN **to distinguish different graph structures**.

- If two different graphs get different representations, the GNN can distinguish them. If they get the same representation for structurally different graphs, the GNN loses some information.

The Weisfeiler–Lehman (WL) Test



WL Test = A method for distinguishing graph structures

- The Weisfeiler–Lehman test repeatedly looks at a node and its neighbourhood and updates its label. It can distinguish many different graph structures.
- The researchers noticed that this process is very similar to what GNNs do.

So they use the **WL test as a benchmark for measuring GNN expressive power**.

The Big Problem: SUM vs MEAN vs MAX

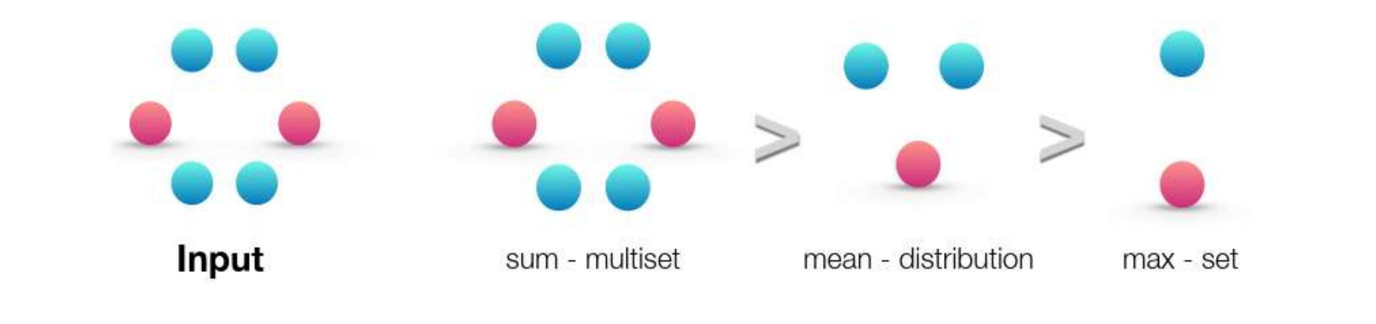


Figure 2: **Ranking by expressive power for sum, mean and max aggregators over a multiset.**

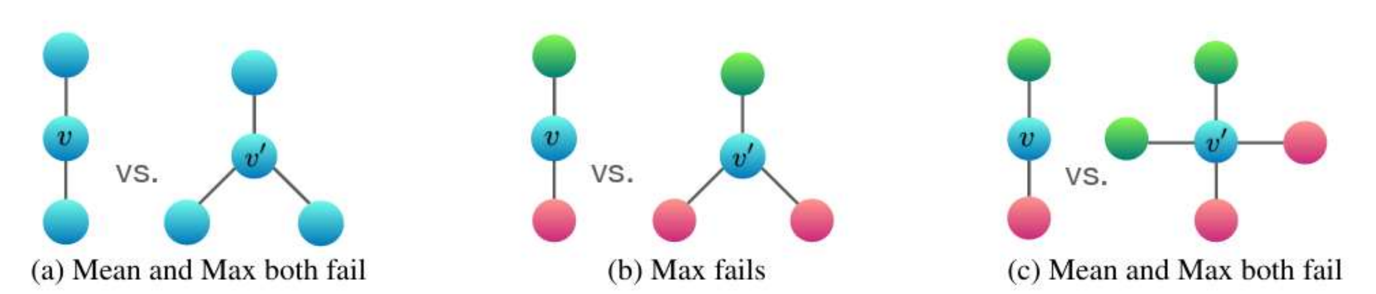


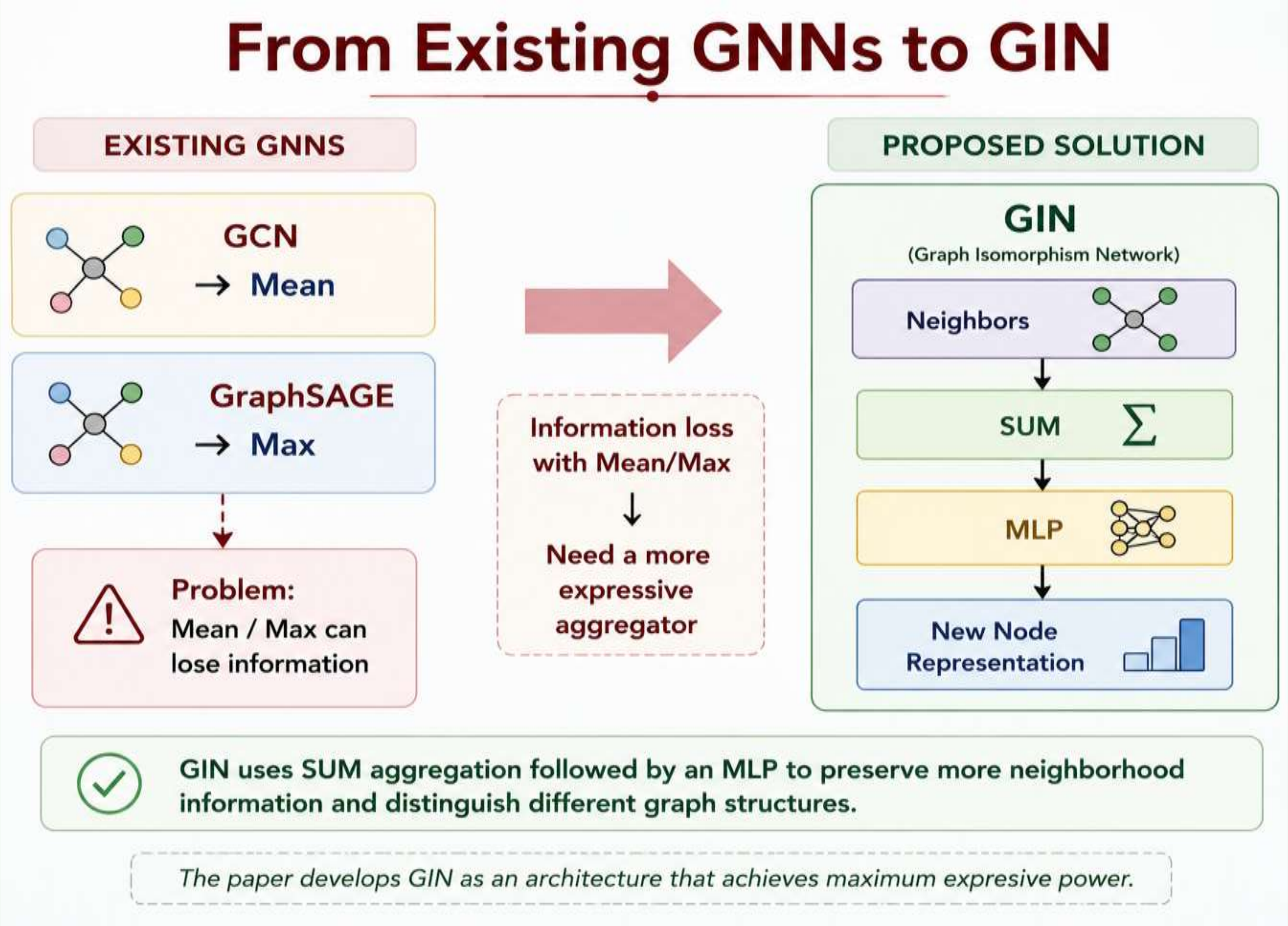
Figure 3: **Examples of graph structures that mean and max aggregators fail to distinguish.**

Aggregation Matters	What it captures
SUM	Full neighbourhood information
MEAN	Distribution / average
MAX	Only strongest/distinct elements

Expressive Power : SUM > MEAN > MAX

For example: **[Red, Red, Blue]**
MAX essentially sees:
[Red, Blue]
It loses the fact that **Red appeared twice**.
The paper demonstrates exactly this limitation.

From existing Graph Neural Network to Graph Isomorphism Network



Main Theoretical Result

1. **GNNs are not at most as powerful as WL**
→ "A GNN cannot distinguish more graph structures than the WL test can."

2. **GIN reaches WL-level expressive power**
→ "GIN can distinguish the same graph structures that the WL test can distinguish."

3. **Therefore, GIN is theoretically highly expressive**
→ "Since GIN reaches the WL test's limit, it is considered highly powerful at capturing graph structure."

Neighbourhood Information
↓
SUM
↓
MLP
↓
GIN
↓
Weis Feiler-Lehman Level Expressive Power

GIN is as powerful as the WL Test

Experiments & Results

- The authors tested GIN against weaker GNN variants on nine datasets. The results supported their theory. For example, on the Reddit-Binary dataset, GIN achieved around 92% accuracy, while the mean-based model was around 50%.

Experimental Evaluation

9 graph classification datasets

Bioinformatics:

- MUTAG
- PTC
- NCI1
- PROTEINS

Social Networks:

- IMDB
- COLLAB
- REDDIT

Model	REDDIT-B
GCN / Mean	**50.0%**
GIN-0	**92.4%**
GIN-ε	**92.2%**

Conclusion

The paper shows:

1. GNNs learn graph structure through neighbourhood aggregation.
2. Aggregation function affects expressive power.
3. Mean and Max can lose structural information.
4. GIN uses **Sum + Multi-Layer Perceptron** to achieve high expressive power.
5. GIN reaches **WL-level expressive power** and performs strongly experimentally.

THANK YOU !